

Math 70

Week 5

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1 Problems

Example 1.1. How many 8 letter words we can create (meaningful or not) using the letters of

”*LAGALAXY*”?

- (A) $8!$ (B) $\frac{8!}{2}$ (C) $\frac{8!}{4}$ (D) $\frac{8!}{6}$ (E) $\frac{8!}{12}$

Example 1.2. You throw a fair coin consequently. What is the probability that HH appears before TT?

Example 1.3. (Final 20 Summer) Balin and Dwalin are two brothers, playing a game with a slightly crooked coin, with $P(H) = 0.6$ and $P(T) = 0.4$. The coin is tossed 10 times (each toss is independent from others) and in any turn it shows heads, it is tossed again. The brothers are counting the cases where the coin is tossed twice and the second toss, too, is heads. Here is an example scenario:

$$T(HT)TTTTTTT(HH)$$

The count here will be 1, as there are only two cases where the coin showed heads and in only one such case, the second toss was also heads. Thus, only the 10th turn is counted.

Let X be the counted number of such double heads. What is the support, pmf, and expected value of X ?

Example 1.4. In an urn there are 6 green, 6 yellow, 6 red, and 6 blue balls.

1. You pick two balls. What is the probability that they are the same color.
2. You pick three colors. What is the probability that you exactly two colors are the same? (Eg BBR, or RRY).

Example 1.5. You throw a dice repeatedly. Let X be the number of throws until you get three 6s. What is the expected number of throws?

Example 1.6. Let $X \sim \text{Pois}(\lambda)$. Compute $E(X)$ and $E(e^{tX})$.

Example 1.7. A *round-robin* tournament is being held with n tennis players; this means that every player will play against every other player exactly once.

1. How many games are played in total?
2. How many possible outcomes are there for the tournament (the outcome lists out who won and who lost for each game)?

Solution